

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

CS 458 - Data Mining (DM)

Fall 2025

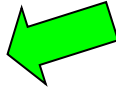
**Lecture 12: Frequent Pattern
Mining**

Mining Frequent Patterns

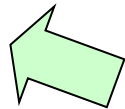
Itemset Mining, Association and Correlation Analysis

Chapter 6 of the Text Book

Chapter 6: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts 
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

Scalable Frequent Itemset Mining Methods

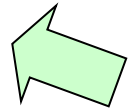
- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori 
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

Further Improvement of the Apriori Method

- Major computational challenges
 - Multiple scans of transaction database
 - Huge number of candidates
 - Tedious workload of support counting for candidates
- Improving Apriori: general ideas
 - Reduce passes of transaction database scans
 - Shrink number of candidates
 - Facilitate support counting of candidates

Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori
- ECLAT: Frequent Pattern Mining with Vertical Data Format



Vertical Data Format

Table 6.1 Transactional Data for an *AllElectronics* Branch

<i>TID</i>	<i>List of item IDs</i>
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

Table 6.3 The Vertical Data Format of the Transaction Data Set *D* of Table 6.1

<i>itemset</i>	<i>TID_set</i>
I1	{T100, T400, T500, T700, T800, T900}
I2	{T100, T200, T300, T400, T600, T800, T900}
I3	{T300, T500, T600, T700, T800, T900}
I4	{T200, T400}
I5	{T100, T800}

T100 I1, I2, I5

T200 I2, I4

T300

T400

Support (I1) = 6

Support(I1,I2) = 4

I1 intersection I2

= {T100, T400, T800, T900}

Table 6.4 2-Itemsets in Vertical Data Format

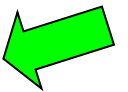
<i>itemset</i>	<i>TID_set</i>
{I1, I2}	{T100, T400, T800, T900}
{I1, I3}	{T500, T700, T800, T900}
{I1, I4}	{T400}
{I1, I5}	{T100, T800}
{I2, I3}	{T300, T600, T800, T900}
{I2, I4}	{T200, T400}
{I2, I5}	{T100, T800}
{I3, I5}	{T800}

Table 6.5 3-Itemsets in Vertical Data Format

<i>itemset</i>	<i>TID_set</i>
{I1, I2, I3}	{T800, T900}
{I1, I2, I5}	{T100, T800}

Chapter 6 Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary



Misleading strong rule

- $N = 10000$
- Games = 6000
- Videos = 7500
- Games and Videos = 4000
- Min_Sup = 30%, Min_Conf = 60%
- Games $\rightarrow$ Videos
 - [support = 40%, Confidence = $P(V / G) = P(V \text{ and } G) / P(G) = 66.7\% = 4000 / 6000$]
- $P(\text{Videos}) = .75 = 7500 / 10000$
- $P(\text{Videos} / \text{Games}) = 0.67$
- Infact computer games and videos are negatively associated
- Lift = ?

	Game	~Game	Sum (row)
Video	4000	3500	7500
~Video	2000	500	2500
Sum(col.)	6000	4000	10000

Interestingness Measure: Correlations (Lift)

- *Buy Games* $\Rightarrow$ *BuyVideos* [40%, 66.7%] is misleading
 - The overall % of video is 75% > 66.7%.
- *Buy Games* $\Rightarrow$ *not videos* [20%, 33.3%] is more accurate, although with lower support and confidence
- Measure of dependent/correlated events: lift

$$lift = \frac{P(A \cup B)}{P(A)P(B)}$$

$$lift(G, V) = \frac{4000/10000}{6000/10000 * 7500/10000} = 0.89$$

$$lift(G, \neg V) = \frac{2000/10000}{6000/10000 * 2500/10000} = 1.33$$

	Game	~Game	Sum (row)
Video	4000	3500	7500
~Video	2000	500	2500
Sum(col.)	6000	4000	10000

$P(A \text{ and } B) = P(A) * P(B)$, when A and B are independent
Lift = $P(A)P(B) / P(A)P(B) = 1$

Basic Concepts: Association Rules

Tid	Items bought
10	cola, Nuts, Diaper
20	cola, Coffee, Diaper
30	cola, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk

$$lift = \frac{P(A \cup B)}{P(A)P(B)}$$

- $cola \rightarrow Diaper$ (60%, 100%)
- $Lift = .6 / (.6)(0.8) = 1.25 > 1$ means positive correlation
- $P(Diaper / Cola) = 100\%$
- $P(Diaper) = 4/5 = 80\%$
- $Diaper \rightarrow cola$ (60%, 75%)

Table 6.6 2×2 Contingency Table Summarizing the Transactions with Respect to Game and Video Purchases

	<i>game</i>	$\overline{\text{game}}$	Σ_{row}
<i>video</i>	4000	3500	7500
$\overline{\text{video}}$	2000	500	2500
Σ_{col}	6000	4000	10,000

Table 6.7 Table 6.6 Contingency Table, Now with the Expected Values

	<i>game</i>	$\overline{\text{game}}$	Σ_{row}
<i>video</i>	4000 (4500)	3500 (3000)	7500
$\overline{\text{video}}$	2000 (1500)	500 (1000)	2500
Σ_{col}	6000	4000	10,000

$$6000 * 7500 / 10000 = 4500$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}} = \frac{(4000 - 4500)^2}{4500} + \frac{(3500 - 3000)^2}{3000} \\ + \frac{(2000 - 1500)^2}{1500} + \frac{(500 - 1000)^2}{1000} = 555.6.$$

Because the χ^2 value is greater than 1, and the observed value of the slot (*game, video*) = 4000, which is less than the expected value of 4500, *buying game* and *buying video* are *negatively correlated*. This is consistent with the conclusion derived from the analysis of the *lift* measure in Example 6.8. ■